

Digital Logic Design

Semester II
FY

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5. Foundations of Digital Design

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5.2 Introduction, realization of logic expressions with PROM, PLA and PAL and their implementation.

5.1 Introduction to Synchronous Machine (Mealy and Moore): Block diagram level. And its representation as state diagram.

INTRODUCTION

- A synchronous state machine is a machine whose transition is controlled by the state signal and occur on the triggering edge of the clock
- A finite state machine is a machine that has many states and has a logical way of changing from one state to the other under guiding rules.
- Two types of synchronous finite state machines to be discussed are Moore and Mealy state machines

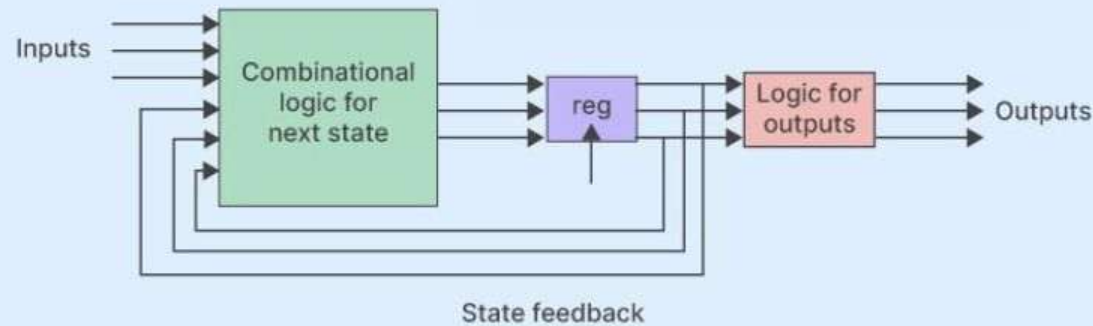
MOORE AND MEALY STATE MACHINES

- A Moore state machine is a machine whose output values are solely dependant on the current state of the machine
- A Mealy machine on the other hand is a state machine whose output values are determined by the current state and the current inputs of the machine

Mealy Machine	Moore Machine
Output depends both upon the present state and the present input	Output depends only upon the present state.
Generally, it has fewer states than Moore Machine.	Generally, it has more states than Mealy Machine.
The value of the output function is a function of the transitions and the changes, when the input logic on the present state is done.	The value of the output function is a function of the current state and the changes at the clock edges, whenever state changes occur.
Mealy machines react faster to inputs. They generally react in the same clock cycle.	In Moore machines, more logic is required to decode the outputs resulting in more circuit delays. They generally react one clock cycle later.

Mealy vs Moore Machine

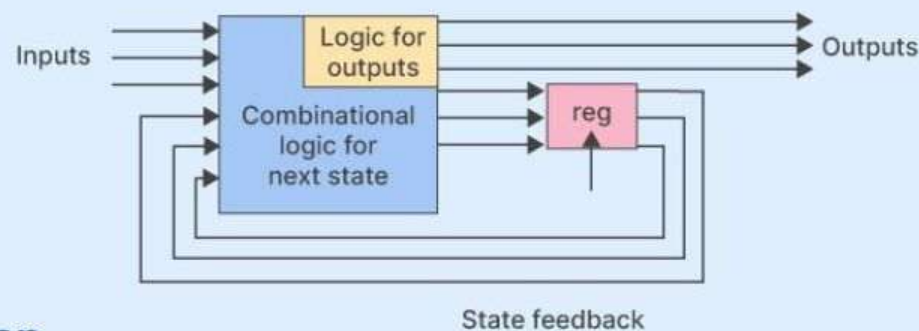
What's the difference?



Moore machine

Outputs are a function of current state

Outputs change synchronously with state changes



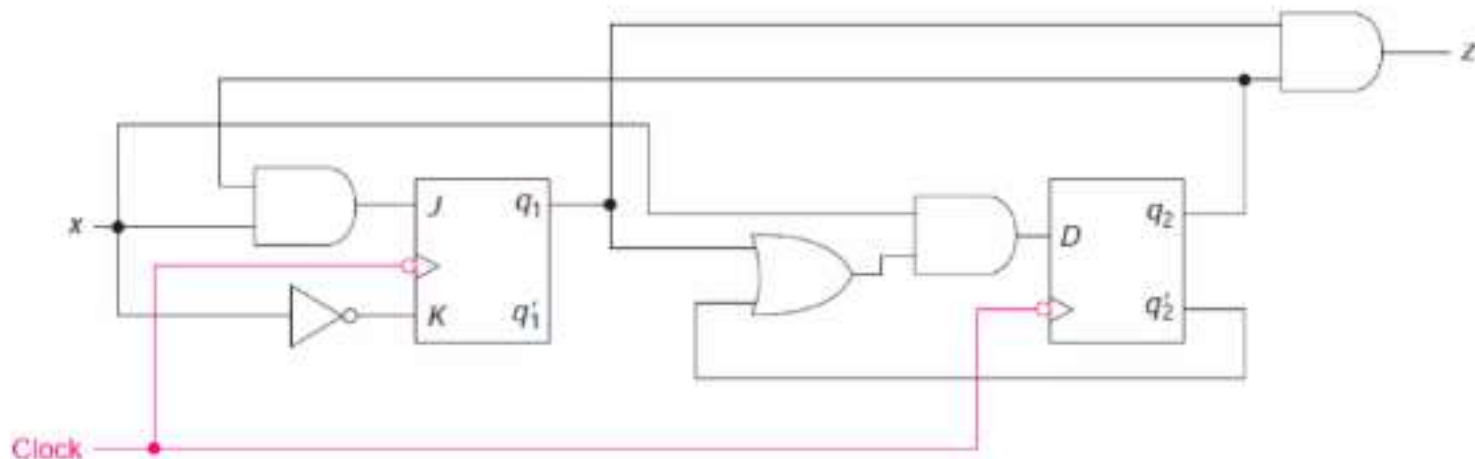
Mealy machine

Outputs depend on state and on inputs

Input changes can cause immediate output changes (asynchronous)

MOORE STATE MACHINE

- The state of a Moore state machine is determined by the state of the machine
- In the diagram below, the state is determined by the state of the JK and D flip-flops



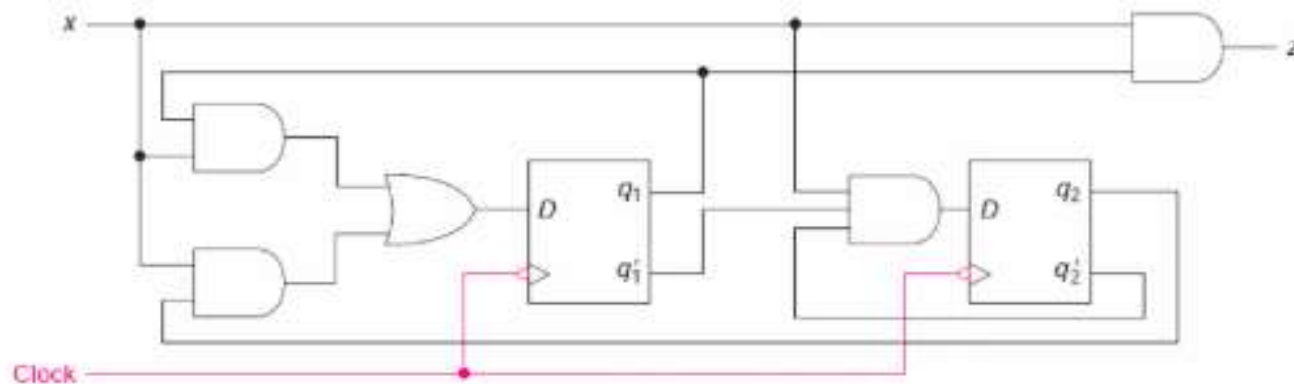
OPERATION OF A MOORE STATE MACHINE

- Below is the state table for the Moore state machine on the previous page
- As seen below, the state of the machine is determined by the state of the machine only

$q_1 q_2$	$q_1^* q_2^*$		z
	$x = 0$	$x = 1$	
0 0	0 0	0 1	0
0 1	0 0	1 0	0
1 0	0 0	1 1	0
1 1	0 0	1 1	1

MEALY STATE MACHINE

- The output depends on the current state and the input
- In the diagram below, the output Z will be determined by the input X and the state of the machine



OPERATION OF A MEALY STATE MACHINE

- As seen below, the output of the mealy state machine are determined by the inputs and the current state of the machine

q	q^*		z	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
0 0	0 0	0 1	0	0
0 1	0 0	1 0	0	0
1 0	0 0	1 0	0	1
1 1	0 0	1 0	0	1

DIFFERENCES BETWEEN MEALY AND MOORE STATE MACHINES

- ❑ The Mealy state machine's state is changed by the inputs and the state of the machine but
- ❑ The Moore state machine's state is affected by the state of the machine only
- ❑ Moore machines are slower than the Mealy machines
- ❑ Moore machines are easier to compose than Mealy machines
- ❑ The Moore state machine takes at least one clock cycle to change the state of the machine while the Mealy state machine is guaranteed to use more than one clock cycle

ADVANTAGES OF MEALY AND MOORE STATE MACHINES

- ❑ Moore machines are cheap
- ❑ They are easy to use
- ❑ Moore state machines are very fast
- ❑ Mealy machines are reactive i.e. they have a low response time (they are fast)

DISADVANTAGES OF MEALY AND MOORE STATE MACHINES

- ❑ Mealy state machines are expensive to produce
- ❑ Number of states can become unmanageable (they become too many)

USES OF MEALY AND MOORE STATE MACHINES

- ❑ Mealy state machines are used in processors due to their property of having many states
- ❑ Mealy state machines are also used to provide a rudimentary mathematical model for cipher machines
- ❑ A Moore state machine is used as a right enable in SRAM because of its speed.
- ❑ It is used in SRAM because SRAM needs a level-sensitive control (signal has to be asserted for an amount of time)

Programmable Logic Array vs Programming Array Logic

Programmable Logic Array (PLA) and Programmable Array Logic (PAL) are both **programmable logic devices (PLDs)** used to implement combinational logic circuits, but they differ mainly in flexibility, speed, and structure.

Programmable Logic Array (PLA)

Both **AND array** and **OR array** are programmable.

Highly **flexible** (can implement any SOP expression)

Allows sharing of product terms

More **complex design**

Slower compared to PAL

More expensive

Inputs → Programmable AND gates → Programmable OR gates → Output

It is efficient when many outputs share common terms

Applications of PLA

1. Custom Reasoning Arrangement: PLAs are used to make exceptionally mechanized reasoning circuits for unequivocal capacities, habitually in embedded structures and purchaser devices.
2. Prototyping: Draftsmen use PLAs to demonstrate and test reasoning plans before zeroing in on ASIC or [FPGA](#) plans.
3. Estimation Execution: They can complete complex computations in hardware, such as encoding/unraveling plans or custom data dealing with computations.
4. Modernized Circuit Unraveling: PLAs develop the arrangement of puzzling high-level circuits by allowing fashioners to design Boolean capacities directly into gear.
5. Control Structures: Used in control systems for programmable state machines and to complete unambiguous control reasoning.

Programmable Array Logic (PAL)

AND array is programmable, OR array is fixed.

Less flexible than PLA

Faster operation

Simpler design

Cheaper and widely used

Inputs → Programmable AND gates → Fixed OR gates → Output

Limited number of product terms per output

PLA → Like a *fully customizable kitchen* (you can design everything)

PAL → Like a *modular kitchen* (some parts are fixed)

Programmable Logic Array vs Programming Array Logic

Feature	PLA	PAL
AND Array	Programmable	Programmable
OR Array	Programmable	Fixed
Flexibility	High	Medium
Speed	Slower	Faster
Cost	Expensive	Cheaper
Complexity	High	Low
Product Terms	Shared among outputs	Fixed per output
Design Ease	Difficult	Easy